
Course Section: Electricity – Monitoring and Measuring a.c.

2022 Ph H2 Q15

(a)

Calculate the peak current

Given:

- $I_{\text{rms}} = 0.65 \text{ A}$

Use:

$$I_{\text{peak}} = \sqrt{2} \times I_{\text{rms}} = 1.414 \times 0.65 = 0.92 \text{ A}$$

(b)

Calculate the energy transferred to the heater in 2.0 minutes

Given:

- $V_{\text{rms}} = 230 \text{ V}$

- $I_{\text{rms}} = 0.65 \text{ A}$

- $t = 2.0 \text{ min} = 120 \text{ s}$

Use:

$$E = IVt = 230 \times 0.65 \times 120 = 17\,940 \text{ J}$$

(c)

State how the energy transferred by a.c. compares with d.c.

✓ The **same amount of energy** is transferred by **a.c. and d.c.** if the **rms values** of voltage and current are used.

✓ The **rms value** of an alternating current is the equivalent d.c. value that would transfer the **same power** to a resistive load.

Revision Tips

- Use $I_{\text{peak}} = \sqrt{2} \times I_{\text{rms}}$ and vice versa
- Always convert minutes to **seconds** when calculating energy or power
- $E = IVt$ is just as valid for a.c. if you use **rms** values
- Emphasise that rms is used **because** it represents equivalent heating effect (power transfer) to d.c.